C 1748 (Pages : 3) Name.....

Reg. No.....

FOURTH SEMESTER M.Sc. DEGREE (SUPPLEMENTARY/IMPROVEMENT) EXAMINATION, MARCH 2021

(CUCSS)

Mathematics

MT 4E 02—ALGEBRAIC NUMBER THEORY

(2016 Admissions)

Time : Three Hours Maximum : 36 Weightage

Part A

Answer **all** *questions.*
Each question carries 1 *weightage.*

1. Define a module. Give an example.
2. Express $t_1^3 + t_2^3 (n = 2)$ in terms of elementary symmetric polynomials.
3. Let $\alpha \in K = Q(\theta)$. Define the field polynomial of α over K.
4. Prove that every quadratic field is of the form $Q(\sqrt{d})$ where $d \in N$ and d is square free.
5. Find an integral basis and discriminant for $Q(\sqrt{3})$.
6. Write all units in $\mathbb{Q}$ and $\mathbb{Z}$.
7. Whether the group of units in $Q(\sqrt{3})$ finite ? Give reasons.
8. Find $\mathfrak{p}^2$ if $\mathfrak{p}$ is the ideal $\langle 2, 1+\sqrt{-5} \rangle$ in $\mathbb{Z}[\sqrt{-5}]$.
9. Find all fractional ideals of $\mathbb{Z}$.
10. State Minkowski's Theorem.
11. Define L^{st} as a vector space over $\mathbb{R}$. What is its dimension ?
12. Define class group and class number of O.
13. Write any six ideals in $\mathbb{Z}[\sqrt{-5}]$ which contains the element 6.
14. State Kummer's special case for Fermat's Last Theorem.

(14 × 1 = 14 weightage)

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Part B

Answer any **seven** *questions.*
Each question has weightage 2.

15. Prove that a subgroup of a finitely generated abelian group is finitely generated.

16. Express $Q\left(\sqrt{3}, \sqrt[3]{5}\right)$ as $Q(\theta)$. Whether the expression unique ? Give reason.

17. Find the minimal polynomial of $\xi = e^{\frac{2\pi i}{p}}$, p an odd prime over Q. Also find the degree of $Q(\xi)$.

18. Prove that the equation $y^2 + 2 = x^3$ has only the integer solutions $y = \pm 5, x = 3$.

19. Let R be a ring and I be an ideal of R. Prove that I is maximal if and only if R/I is a field.

20. Prove that the factorization of elements of O into irreducibles is unique if and only if every ideal of O is principal.

21. Prove that the class group of a number field is a finite abelian group and the class number h is finite.

22. Which of the following elements in $\mathbb{Z}[i]$ are irreducible ?

    $1+i, 3-7i, 5, 7, 12i, -4+5i$.

23. Prove that $x^4 + y^4 = z^2$ has no integer solutions with $x, y, z \neq 0$.

24. Prove that for each $\alpha \in \mathbb{Z}[\xi]$ there exists $\alpha \in \mathbb{Z}$ such that $\alpha^p \equiv a \left(\text{mod } I^p\right)$ where $I = (1-\xi)$..

($7 \times 2 = 14$ weightage)

Part C

Answer any **two** *questions.*
Each question has weightage 4.

25. (a) Prove that every subgroup of a free abelian group of rank n is free of rank $s \leq n$.

    (b) Prove that an algebraic number α is an algebraic integer if and only if its minimum polynomial over Q has coefficients in $\mathbb{Z}$.

26. (a) Prove that the ring of integers of $\mathbb{Q}(\xi)$ is $\mathbb{Z}(\xi)$.

    (b) Write the group of units of the ring of integers of $Q\left(\sqrt{-1}\right), Q\left(\sqrt{-2}\right), Q\left(\sqrt{-3}\right), Q\left(\sqrt{-5}\right), Q\left(\sqrt{-6}\right)$.

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27. (a) Prove that every non-zero ideal of O can be written as a product of prime ideals uniquely upto the order of factors.

(b) In an integral domain D, show that a principal ideal $\langle p \rangle$ is prime if and only if p is a prime or zero.

28. (a) Prove that a complex number θ is an algebraic integer if and only if the additive group generated by $1, \theta, \theta^2, \theta^3, \ldots$ is finitely generated.

(b) If $[K : \mathbb{Q}] = n$ and a $\alpha \in \mathbb{Q}$ then show that $N_K(\alpha) = \alpha^n$ *and* $T_K(\alpha) = n\alpha$.

(2 × 4 = 8 weightage)

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